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Section 03

CSE 422 (RRH)

Assignment 02

① Maximum Mass = 13kg

a) Population:

Chromosome 1: 1101

Chromosome 2: 0110

Chromosome 3: 1010

Chromosome 4: 0011

b) Fitness: If total mass ≤ 13 , fitness = total price

Chromosome 1: $20 + 50 + 30 = 100$

Chromosome 2: $50 + 10 = 60$

Chromosome 3: $20 + 10 = 30$

Chromosome 4: $10 + 30 = 40$

Two fittest chromosome = chromosome 1 and 2

c) Crossover:

Chromosome 1: 110|1 $\swarrow \searrow$ 0111
Chromosome 2: 011|0 $\swarrow \searrow$ 1100

Offspring 1: 0111

Offspring 2: 1100

d) Mutation: Offspring 1: 0101

Fitness:

Offspring 1: 80

Offspring 2: 70

② a) Max weight = 10 kg

chromosome 1 : 10100001

chromosome 2 : 01000101

chromosome 3 : 00000011

chromosome 4 : 00001001

b) If total weight ≤ 10 , fitness = total reward

chromosome 1 : $20 + 10 + 7 = 37$

chromosome 2 : $5 + 25 + 7 = 37$

chromosome 3 : $4 + 7 = 11$

chromosome 4 : $15 + 7 = 22$

fittest chromosome : chromosome 1 and 2

c) Crossover :

chromosome 1 : 1010|0001 $\swarrow$ 01000001
chromosome 2 : 0100|0101 $\searrow$ 10100101

offspring 1 : 01000001

offspring 2 : 10100101

d) Mutation :

offspring 2 : 00100101

fitness :

offspring 1 = $5 + 7 = 12$

offspring 2 = $10 + 25 + 7 = 42$

③

Chromosome 1: 5, 12, 34, 23, 45, 67, 89, 30, 16, 7

Chromosome 2: 2, 3, 5, 7, 11, 13, 15, 19, 23, 29

Chromosome 3: 67, 61, 59, 41, 17, 36, 34, 28, 25, 24

Chromosome 4: 37, 59, 61, 31, 4, 8, 9, 10, 25, 28

Fitness: $||P - C| - 100|$

Chromosome 1:

$$P = 5 + 23 + 67 + 89 + 7 = 191$$

$$C = 12 + 34 + 45 + 30 + 16 = 137$$

$$\text{Fitness} = ||P - C| - 100| = 46$$

Chromosome 2:

$$P = 2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 + 23 + 29 = 129$$

$$C = 0$$

$$\text{Fitness} = ||P - C| - 100| = |129 - 100| = 29$$

Chromosome 3:

$$P = 67 + 61 + 59 + 41 + 17 = 245$$

$$C = 36 + 34 + 28 + 25 + 24 = 147$$

$$\therefore \text{Fitness} = ||P - C| - 100| = 2$$

Chromosome 4:

$$P = 37 + 59 + 61 + 31 = 188$$

$$C = 4 + 8 + 9 + 10 + 25 + 28 = 84$$

$$\text{Fitness} = ||P - C| - 100| = 4$$

Selection: chromosome 3, 4

Crossover :

chromosome 3: 67, 61, 59, 41, 17, 36, 34, 28, 25, 24

chromosome 4: 37, 59, 61, 31, 4, 8, 9, 10, 25, 28

offspring 1: 67, 61, 59, 41, 17, 36, 34, 28, 25, 28

offspring 2: 37, 59, 61, 31, 4, 8, 9, 10, 25, 24

Mutation :

offspring 1: 67, 61, 59, 41, 17, 36, 34, 28, 25, 22

Evaluate :

Fitness of offspring 1 :

$$P = 67 + 61 + 59 + 41 + 17 = 245$$

$$C = 36 + 34 + 28 + 25 + 22 = 145$$

$$\text{Fitness} = ||245 - 145| - 100| = 0$$

Fitness of offspring 2 :

$$P = 37 + 59 + 61 + 31 = 188$$

$$C = 4 + 8 + 9 + 10 + 25 + 24 = 80$$

$$\text{Fitness} = ||188 - 80| - 100|$$

$$= 8$$

④

Chromosome 1 : 7, 4, 6, 5, 3, 5

Chromosome 2 : 1, 8, 6, 5, 9, 1

Chromosome 3 : 9, 1, 1, 2, 3, 4

Chromosome 4 : 5, 5, 4, 7, 6, 3

Fitness:

Chromosome 1:

Sum = 30

2 adj not equal

even = 2

Start with odd

fitness = 4

Chromosome 2:

Sum = 30

2 adj not equal

even = 2

Start with odd

fitness = 4

Chromosome 3:

Sum = 20

2 adj equal

even = 2

Start with odd

fitness = 2

Chromosome 4 :

Sum = 30

2 adj equal

even = 2

Start with odd

Fitness = 3

Select : chromosome 1 and 2

Crossover :

Chromosome 1 : 7, 4, 6, 5, 3, 5

Chromosome 2 : 1, 8, 6, 5, 9, 1

Offspring 1 : 7, 4, 6, 5, 9, 1

Offspring 2 : 1, 8, 6, 5, 3, 5

Mutation : Offspring 2 : 1, 8, 6, 7, 3, 5

Fitness :

Offspring 1 :

Sum = 32

2 adj not equal

even = 2

Start with odd

Fitness = 3

Offspring 2 :

Sum = 30

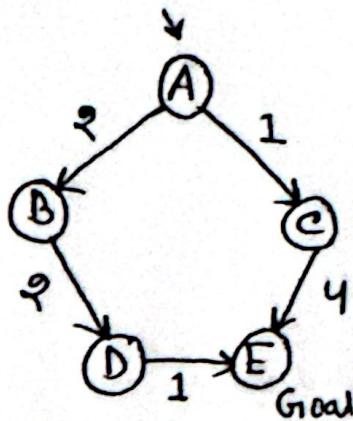
2 adj not equal

even = 2

Start with odd

Fitness = 4

⑤



$$h(A) = 4$$

$$h(B) = 2$$

$$h(C) = 1$$

$$h(D) = 1$$

$$h(E) = 0$$

Consistency check:

$$h(n) \leq c(n, n') + h(n')$$

$$h(A) \leq c(A, B) + h(B)$$

$$4 \leq 2 + 2 = 4$$

$$h(A) \leq c(A, C) + h(C)$$

$$4 \leq 1 + 1 = 2 ; \text{ not consistent}$$

Admissibility check:

$$h(n) \leq h^*(n)$$

$$h(A) \leq h^*(A)$$

$$4 \leq 5$$

$$h(B) \leq h^*(B)$$

$$2 \leq 3$$

$$h(C) \leq h^*(C)$$

$$1 \leq 4$$

$$h(D) \leq h^*(D)$$

$$1 \leq 1$$

$$h(E) \leq h^*(E)$$

$$0 \leq 0$$

So the graph is admissible but inconsistent.

⑥ The hill climbing algorithm is a local search method that continuously moves in the direction of increasing value to find the best solution. While simple and fast, it suffers from several issues, especially in complex search spaces like the 8-Queens problem.

Issues with the Hill-Climbing algorithm:

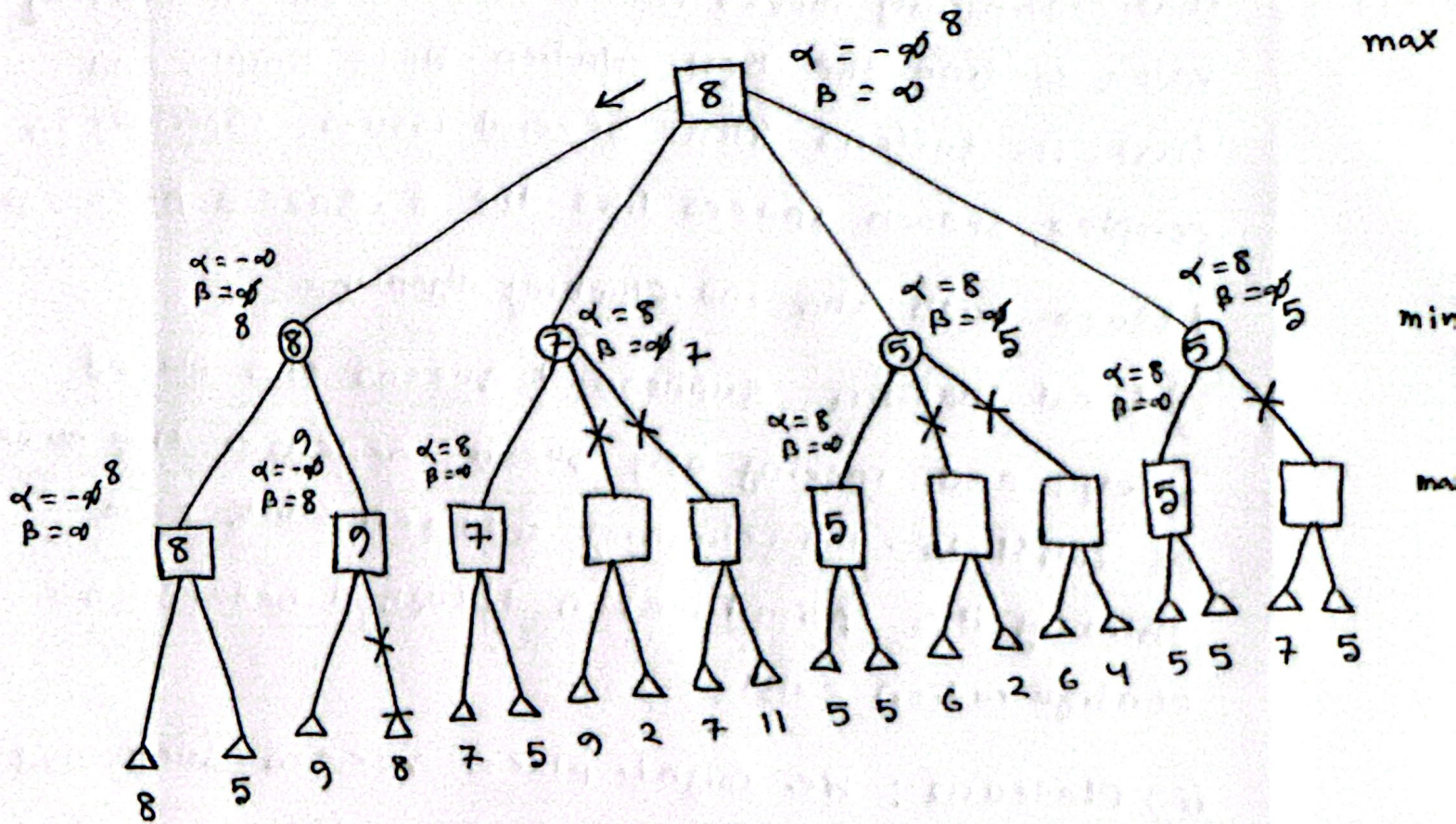
i) Local maxima: Suppose 7 queens are placed safely, and moving any queen increases the number of conflicts. Hill-climbing will get stuck here thinking it's optimal, even though a better global configuration exists.

ii) Plateaus: We might place queens such that many board configurations have the same number of conflicts and no move gives improvement. The algorithm will get confused and immediately stop there.

iii) Ridges: We may need to temporarily increase the number of conflicts to eventually reach a better configuration - but hill climbing avoids that.

max = α
min = β

⑦



⑧

